

Homework 4: Markov models for sequence alignment

Bernhard Buss Dario Franzen Aurimas Greicius
 Timothy Lauren Nijhuis Carlo Rossetto Yuliyana Slavchev
 Gökem Kadir Solun

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Problem 8: Estimating emission and transition probabilities for profile HMM

From Figure 1, the match columns are 1, 2, 3 and the insert region is the contiguous block between match positions 2 and 3. The five aligned sequences are:

-	M_1	M_2	I	I	I	M_3
bat	A	G	—	—	—	C
rat	A	—	A	G	—	C
cat	A	G	—	—	—	C
gnat	—	G	A	A	A	C
goat	A	G	—	—	A	C

1. Match emission probabilities

Using the add-one estimate

$$E_i(a) = \frac{N_i(a) + 1}{\sum_{a' \in \{A, C, G, T\}} (N_i(a') + 1)}$$

we obtain:

- **Position 1:** Observed symbols: $A, A, A, -, A$, so $N_1(A) = 4$ and $N_1(C) = N_1(G) = N_1(T) = 0$.
 – Hence, $E_1(A) = \frac{5}{8}$ and $E_1(C) = E_1(G) = E_1(T) = \frac{1}{8}$.

- **Position 2:** Observed symbols: $G, -, G, G, G$, so $N_2(G) = 4$ and $N_2(A) = N_2(C) = N_2(T) = 0$.
 – Hence, $E_2(G) = \frac{5}{8}$ and $E_2(A) = E_2(C) = E_2(T) = \frac{1}{8}$.
- **Position 3:** Observed symbols: C, C, C, C, C , so $N_3(C) = 5$ and $N_3(A) = N_3(G) = N_3(T) = 0$.
 – Hence, $E_3(C) = \frac{6}{9} = \frac{2}{3}$ and $E_3(A) = E_3(G) = E_3(T) = \frac{1}{9}$.

Therefore,

$$E_1 = (A : \frac{5}{8}, C : \frac{1}{8}, G : \frac{1}{8}, T : \frac{1}{8})$$

$$E_2 = (A : \frac{1}{8}, C : \frac{1}{8}, G : \frac{5}{8}, T : \frac{1}{8})$$

$$E_3 = (A : \frac{1}{9}, C : \frac{2}{3}, G : \frac{1}{9}, T : \frac{1}{9})$$

2. Insert emission probabilities

All three insert columns belong to the same insert state I_2 . Ignoring gaps, the insert emissions are:

rat: A, G

gnat: A, A, A

goat: A

So the counts are $N_{I_2}(A) = 5$, $N_{I_2}(G) = 1$, and $N_{I_2}(C) = N_{I_2}(T) = 0$.

Using the same add-one smoothing, we get:

$$E_{I_2}(a) = \frac{N_{I_2}(a) + 1}{\sum_{a' \in \{A, C, G, T\}} (N_{I_2}(a') + 1)}$$

Thus, $E_{I_2}(A) = \frac{6}{10} = \frac{3}{5}$, $E_{I_2}(G) = \frac{2}{10} = \frac{1}{5}$, and $E_{I_2}(C) = E_{I_2}(T) = \frac{1}{10}$.

$$E_{I_2} = (A : \frac{3}{5}, C : \frac{1}{10}, G : \frac{1}{5}, T : \frac{1}{10})$$

For the other insert states I_0, I_1, I_3 there are no observed insert emissions in the alignment, so with add-one smoothing they are uniform:

$$E_{I_0} = E_{I_1} = E_{I_3} = (A : \frac{1}{4}, C : \frac{1}{4}, G : \frac{1}{4}, T : \frac{1}{4})$$

3. Transition probabilities

We use:

$$T_i(k \rightarrow l) = \frac{N_i(k \rightarrow l) + 1}{\sum_{l' \in \{M, I, D\}} (N_i(k \rightarrow l') + 1)}.$$

Observed state paths

The five sequences correspond to:

bat: $M_0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow M_4$

rat: $M_0 \rightarrow M_1 \rightarrow D_2 \rightarrow I_2 \rightarrow I_2 \rightarrow M_3 \rightarrow M_4$

cat: $M_0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow M_4$

gnat: $M_0 \rightarrow D_1 \rightarrow M_2 \rightarrow I_2 \rightarrow I_2 \rightarrow I_2 \rightarrow M_3 \rightarrow M_4$

goat: $M_0 \rightarrow M_1 \rightarrow M_2 \rightarrow I_2 \rightarrow M_3 \rightarrow M_4$

Position $i = 0$; from Begin (M_0)

Counts:

$$N_0(M \rightarrow M) = 4, \quad N_0(M \rightarrow I) = 0, \quad N_0(M \rightarrow D) = 1$$

Therefore:

$$T_0(M \rightarrow M) = \frac{5}{8}, \quad T_0(M \rightarrow I) = \frac{1}{8}, \quad T_0(M \rightarrow D) = \frac{1}{4}$$

Position $i = 1$; from states at position 1 to position 2. From M_1 :

$$N_1(M \rightarrow M) = 3, \quad N_1(M \rightarrow I) = 0, \quad N_1(M \rightarrow D) = 1$$

So

$$T_1(M \rightarrow M) = \frac{4}{7}, \quad T_1(M \rightarrow I) = \frac{1}{7}, \quad T_1(M \rightarrow D) = \frac{2}{7}$$

From D_1 :

$$N_1(D \rightarrow M) = 1, \quad N_1(D \rightarrow I) = 0, \quad N_1(D \rightarrow D) = 0$$

So,

$$T_1(D \rightarrow M) = \frac{1}{2}, \quad T_1(D \rightarrow I) = \frac{1}{4}, \quad T_1(D \rightarrow D) = \frac{1}{4}$$

There are no observed transitions from I_1 , so with add-one smoothing:

$$T_1(I \rightarrow M) = T_1(I \rightarrow I) = T_1(I \rightarrow D) = \frac{1}{3}$$

Position $i = 2$; from states at position 2 to position 3. From M_2 :

$$N_2(M \rightarrow M) = 2, \quad N_2(M \rightarrow I) = 2, \quad N_2(M \rightarrow D) = 0$$

So,

$$T_2(M \rightarrow M) = \frac{3}{7}, \quad T_2(M \rightarrow I) = \frac{3}{7}, \quad T_2(M \rightarrow D) = \frac{1}{7}$$

From D_2 :

$$N_2(D \rightarrow M) = 0, \quad N_2(D \rightarrow I) = 1, \quad N_2(D \rightarrow D) = 0$$

So,

$$T_2(D \rightarrow M) = \frac{1}{4}, \quad T_2(D \rightarrow I) = \frac{1}{2}, \quad T_2(D \rightarrow D) = \frac{1}{4}$$

From I_2 :

$$N_2(I \rightarrow M) = 3, \quad N_2(I \rightarrow I) = 3, \quad N_2(I \rightarrow D) = 0$$

since there are three exits from I_2 to M_3 and three self-loops in I_2 .

Hence,

$$T_2(I \rightarrow M) = \frac{4}{9}, \quad T_2(I \rightarrow I) = \frac{4}{9}, \quad T_2(I \rightarrow D) = \frac{1}{9}$$

Position $i = 3$: from states at position 3 to End (M_4)

All observed paths go from M_3 to End, so

$$N_3(M \rightarrow M) = 5, \quad N_3(M \rightarrow I) = 0, \quad N_3(M \rightarrow D) = 0$$

and thus,

$$T_3(M \rightarrow M) = \frac{3}{4}, \quad T_3(M \rightarrow I) = \frac{1}{8}, \quad T_3(M \rightarrow D) = \frac{1}{8}$$

There are no observed transitions from D_3 or I_3 , so with add-one smoothing:

$$T_3(D \rightarrow M) = T_3(D \rightarrow I) = T_3(D \rightarrow D) = \frac{1}{3}$$

$$T_3(I \rightarrow M) = T_3(I \rightarrow I) = T_3(I \rightarrow D) = \frac{1}{3}$$

A further remark is that from position 3, the following transitions: $T_3(D \rightarrow D)$, $T_3(I \rightarrow D)$, $T_3(M \rightarrow D)$ are not only not observed, but also not feasible, given D_4 and I_4 are not present in the proposed HMM topology in Figure 1 in the instructions. Probabilities are assigned to them as per the formula in the instructions provided.

Summary tables for the transitions probabilities are provided in fraction and decimal format below.

	T_1	T_2	T_3
$M \rightarrow M$	$\frac{4}{7}$	$\frac{1}{4}$	$\frac{3}{4}$
$M \rightarrow I$	$\frac{1}{7}$	$\frac{1}{4}$	$\frac{1}{8}$
$M \rightarrow D$	$\frac{2}{7}$	$\frac{1}{4}$	$\frac{1}{8}$
$D \rightarrow M$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{3}$
$D \rightarrow I$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{3}$
$D \rightarrow D$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{3}$
$I \rightarrow M$	$\frac{1}{3}$	$\frac{4}{9}$	$\frac{1}{3}$
$I \rightarrow I$	$\frac{1}{3}$	$\frac{4}{9}$	$\frac{1}{3}$
$I \rightarrow D$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{3}$

	T_1	T_2	T_3
$M \rightarrow M$	0.571	0.429	0.750
$M \rightarrow I$	0.143	0.429	0.125
$M \rightarrow D$	0.286	0.143	0.125
$D \rightarrow M$	0.500	0.250	0.333
$D \rightarrow I$	0.250	0.500	0.333
$D \rightarrow D$	0.250	0.250	0.333
$I \rightarrow M$	0.333	0.444	0.333
$I \rightarrow I$	0.333	0.444	0.333
$I \rightarrow D$	0.333	0.111	0.333

Problem 9: Protein family membership classification

```
source("code/profileHMM.R")

library(ggplot2)
library(patchwork)
library(knitr)
```

1) Read the two alignments 'GTP_binding_proteins.txt' and 'ATPases.txt' into memory using the function `parseAlignment()` from "profileHMM.R"

```
# Read the two alignments
setwd("~/My Documents/Carlo/ETH UBas UZH - MSc CBB/636-0702-00L Statistical Models in Computa
GTP_binding_proteins <- parseAlignment("data/GTP_binding_proteins.txt")
ATPases <- parseAlignment("data/ATPases.txt")
```

2) Use the function `learnHMM()` from "profileHMM.R" to parametrise two profile HMMs: one for each protein family (multiple alignment)

```
parGTP <- learnHMM(GTP_binding_proteins)
parATP <- learnHMM(ATPases)
```

3) For each family, identify the position(s) with the highest match and with the highest insert emission frequencies over all symbols. Plot the respective match and insert emission frequencies for the identified positions

```
# Identify which col has the highest freq
maxMatchGTP <- which(parGTP$mE == max(parGTP$mE, na.rm = TRUE),
                     arr.ind = TRUE)
maxInsertGTP <- which(parGTP$iE == max(parGTP$iE, na.rm = TRUE),
                     arr.ind = TRUE)

maxMatchATP <- which(parATP$mE == max(parATP$mE, na.rm = TRUE),
                     arr.ind = TRUE)
maxInsertATP <- which(parATP$iE == max(parATP$iE, na.rm = TRUE),
                     arr.ind = TRUE)
```

```

# Collect data in one data frame for GTP
df_GTP <- data.frame(
  Amino = rownames(parGTP$mE),
  MatchProb = parGTP$mE[, maxMatchGTP[2]],
  InsertProb = parGTP$iE[, maxInsertGTP[2]]
)

# Create plots for match and insertion
plotMatchGTP <- ggplot(df_GTP, aes(x = MatchProb, y = Amino)) +
  geom_col() +
  theme_minimal() +
  labs(title = paste("Match state at Position", maxMatchGTP[2]),
       x = "Frequency",
       y = "Amino Acid") +
  scale_y_discrete(limits=rev)

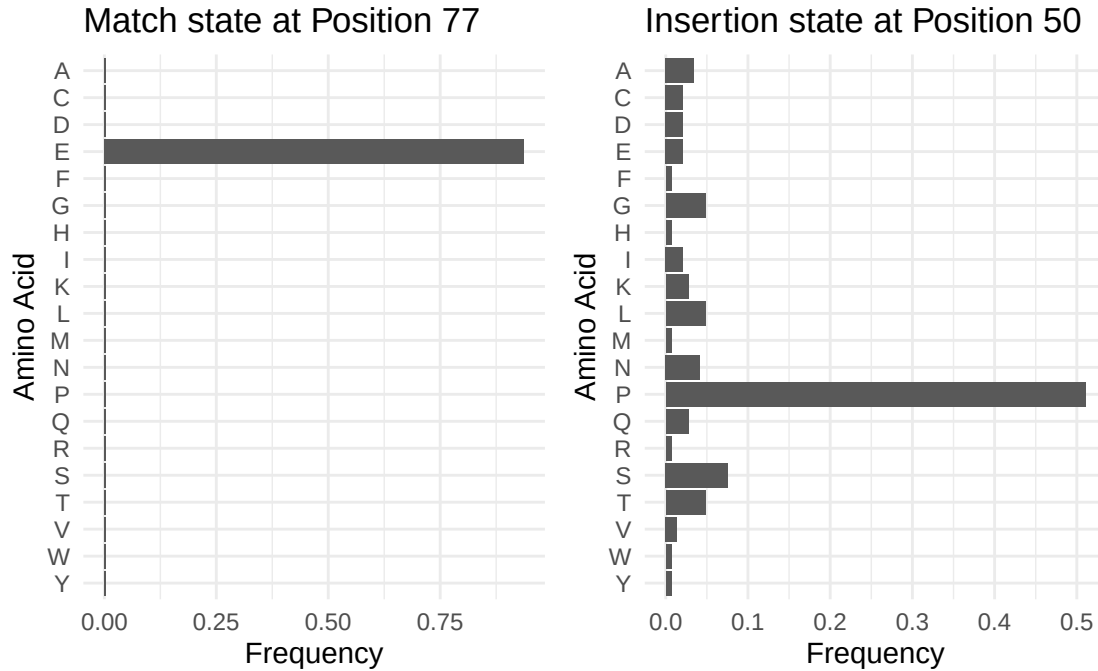
plotInsertGTP <- ggplot(df_GTP, aes(x = InsertProb, y = Amino)) +
  geom_col() +
  theme_minimal() +
  labs(title = paste("Insertion state at Position", maxInsertGTP[2]),
       x = "Frequency",
       y = "Amino Acid") +
  scale_y_discrete(limits=rev)

# Combine into one plot
combinedGTP <- plotMatchGTP + plotInsertGTP +
  plot_annotation(
    title = "GTP Family: Maximum Emission Frequencies",
    theme = theme(plot.title = element_text(face = "bold", hjust = 0.5))
  )

combinedGTP

```

GTP Family: Maximum Emission Frequencies



```
# Collect data in one data frame for ATP
df_ATP <- data.frame(
  Amino = rownames(parATP$mE),
  MatchProb = parATP$mE[, maxMatchATP[2]],
  InsertProb = parATP$iE[, maxInsertATP[2]]
)

# Create plots for match and insertion
plotMatchATP <- ggplot(df_ATP, aes(x = MatchProb, y = Amino)) +
  geom_col() +
  theme_minimal() +
  labs(title = paste("Match state at Position", maxMatchATP[2]),
       x = "Frequency",
       y = "Amino Acid") +
  scale_y_discrete(limits=rev)

plotInsertATP <- ggplot(df_ATP, aes(x = InsertProb, y = Amino)) +
  geom_col() +
  theme_minimal() +
  labs(title = paste("Insertion state at Position", maxInsertATP[2]),
```

```

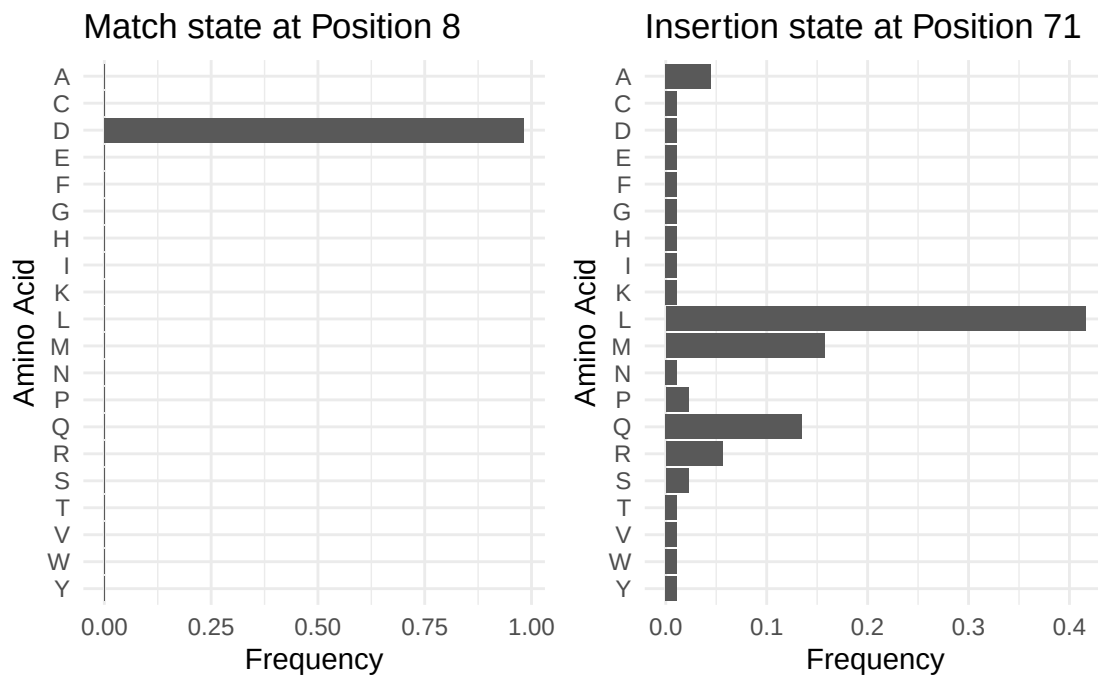
    x = "Frequency",
    y = "Amino Acid") +
  scale_y_discrete(limits=rev)

# Combine into one plot
combinedATP <- plotMatchATP + plotInsertATP +
  plot_annotation(
    title = "ATP Family: Maximum Emission Frequencies",
    theme = theme(plot.title = element_text(face = "bold", hjust = 0.5))
  )

combinedATP

```

ATP Family: Maximum Emission Frequencies



4) Complete the forward() function by filling out the recursion step.

```

#
# The Forward algorithm
#

```

```

forward <- function(HMM, seq) {
  L <- HMM$L
  mE <- HMM$mE
  iE <- HMM$iE
  T <- HMM$T
  len <- length(seq)

  # Random model: same emission probability for all letters in the alphabet
  qr <- 1/nrow(mE)
  # Initialize score matrices for match, insert and delete
  Fm<- Fi<- Fd<- matrix(-Inf, nrow=len+1, ncol=L+1)
  # First column for the begin state (M0) (indices get shifted by 1)
  Fm[1,1]<-0

  # We then allow transitions to I0 and D1 (start with insertion or deletion)
  for(i in 2:len+1){
    Fi[i,1] <- log(iE[seq[i-1],1]/qr) +
      log(T['MI',1]*exp(Fm[i-1,1]) +
        T['II',1]*exp(Fi[i-1,1]))
  }

  # i is the pos in the sequence, j is the profile state
  for(i in 2:(len+1)) {
    for(j in 2:(L+1)) {
      ##### your code here to compute
      #Fm[i, j], Fi[i,j], and Fd[i,j]

      # Calculate the score for the match state
      Fm[i,j] = log(mE[seq[i-1],j]/qr) + log(
        T["MM", j-1] * exp(Fm[i-1, j-1])
        + T["IM", j-1] * exp(Fi[i-1, j-1])
        + T["DM", j-1] * exp(Fd[i-1, j-1])
      )

      # Calculate the score for the insertion state
      Fi[i,j] = log(iE[seq[i-1],j]/qr) + log(
        T["MI", j] * exp(Fm[i-1, j])
        + T["II", j] * exp(Fi[i-1, j])
        + T["DI", j] * exp(Fd[i-1, j])
      )

      # Calculate the score for the deletion state

```

```

    Fd[i,j] = log(T["MD", j-1] * exp(Fm[i, j-1])
                + T["ID", j-1] * exp(Fi[i, j-1])
                + T["DD", j-1] * exp(Fd[i, j-1]))

    ##### your code ends
  }
}

# Termination step: Fe = Fm[len,L+2] = log(P(x|M)/P(x|R))
Fe <- log(T['MM',L+1]*exp(Fm[len+1,L+1]) +
         T['IM',L+1]*exp(Fi[len+1,L+1]) +
         T['DM',L+1]*exp(Fd[len+1,L+1]))
return(Fe)
}

```

5) The file `Unclassified_proteins.txt` contains 31 protein sequences from unknown families. Load the protein sequences into a list using the `parseProteins()` function from in “`profileHMM.R`”.

```
proteinList <- parseProteins("data/Unclassified_proteins.txt")
```

6) For each unclassified protein $x^{(i)}$ in the list, apply the forward algorithm for both models M_1 and M_2 to obtain the log odds ratio. Plot the values $q(x^i)$. Which proteins in the list belong to which family? Can you clearly decide for each protein?

```

# Iterate over the protein sequences and calculate the log odds ratio
logOdds <- c(rep(0, length(proteinList)))

for (i in seq_along(proteinList)) {
  logOddsGTP <- forward(parGTP, proteinList[[i]])
  logOddsATP <- forward(parATP, proteinList[[i]])

  # Subtract to get the ratio since log
  logOdds[i] <- logOddsGTP - logOddsATP
}

```

```

# Plot the ratios
df_ratio <- data.frame(

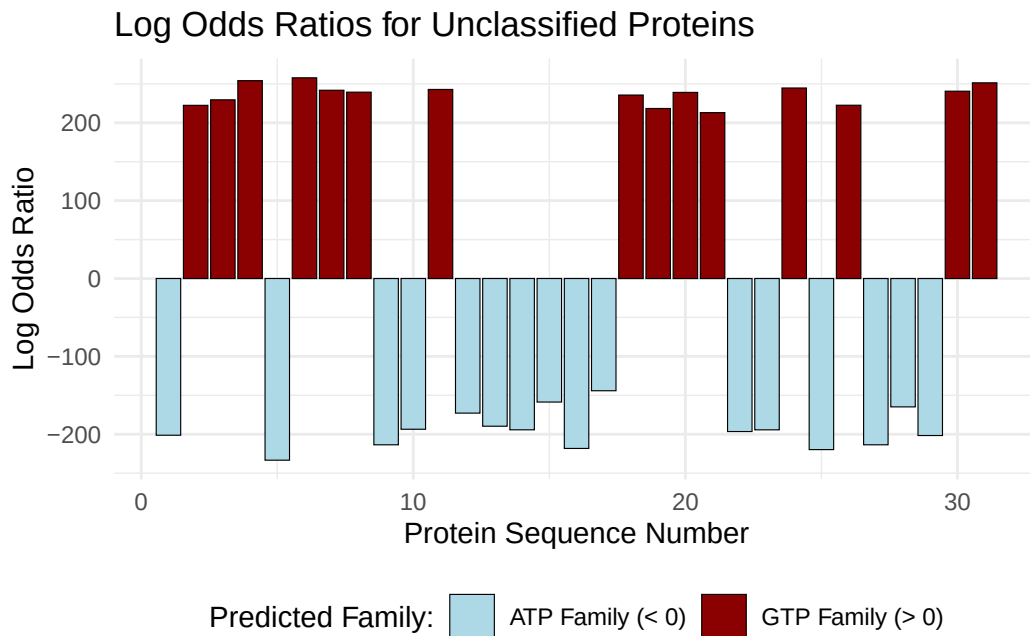
```

```

seqNum = seq(1:length(proteinList)),
ratio = logOdds
)

ggplot(df_ratio, aes(x = seqNum, y = ratio, fill = ratio > 0)) +
  geom_col(color = "black", linewidth = 0.2) +
  theme_minimal() +
  scale_fill_manual(
    name = "Predicted Family:",
    values = c("TRUE" = "darkred", "FALSE" = "lightblue"),
    labels = c("TRUE" = "GTP Family (> 0)", "FALSE" = "ATP Family (< 0)")
  ) +
  labs(
    title = "Log Odds Ratios for Unclassified Proteins",
    x = "Protein Sequence Number",
    y = "Log Odds Ratio"
  ) +
  theme(legend.position = "bottom")

```



```

# Create a table for each family predicition
df_ratio$Family <- ifelse(df_ratio$ratio > 0, "GTP", "ATP")
df_ratio$Score <- round(df_ratio$ratio, 3)

```

```

gtp_proteins <- df_ratio[df_ratio$Family == "GTP", c("seqNum", "Score")]
atp_proteins <- df_ratio[df_ratio$Family == "ATP", c("seqNum", "Score")]

# GTP Table
kable(gtp_proteins,
      col.names = c("Sequence number", "Log-Odds score"),
      align = "c",
      caption = "Proteins classified as GTP family, positive score",
      row.names = FALSE)

```

Table 4: Proteins classified as GTP family, positive score

Sequence number	Log-Odds score
2	222.448
3	229.542
4	254.115
6	257.799
7	241.813
8	239.352
11	242.864
18	235.651
19	218.396
20	239.051
21	213.103
24	244.696
26	222.585
30	240.588
31	251.338

```

# ATP Table
kable(atp_proteins,
      col.names = c("Sequence number", "Log-Odds score"),
      align = "c",
      caption = "Proteins classified as ATP family, negative score",
      row.names = FALSE)

```

Table 5: Proteins classified as ATP family, negative score

Sequence number	Log-Odds score
1	-201.276
5	-233.324
9	-213.540
10	-193.592
12	-172.862
13	-189.699
14	-194.320
15	-158.669
16	-218.264
17	-144.132
22	-196.524
23	-194.348
25	-219.736
27	-213.547
28	-164.847
29	-201.635

The prediction for each protein is visible in the tables above. The decision of which protein belongs to which family is clear in this case, since the log-odds ratio, $q(x^{(i)})$, closest to zero across both families (GTP: 213.103, ATP: -144.132) remains far from 0, indicating confident classification for all proteins.